

Recursive Science

What repeats and reinjected papers added to a 10-by-10 Pathfinder experiment

Pathfinder experiment record

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Abstract

Can a research pipeline use its own completed papers as productive inputs? We examine a Pathfinder experiment beginning with ten papers returned for “mechanism design” and ten for “agentic cooperation”. After investigating 14 selected pairs and repeating those investigations, we compared another 14-pair repeat with 14 pairings of original papers and previously accepted manuscripts. All 28 pilot pipelines finished. The repeat arm produced seven internally accepted papers and reinjection three, but claim-level comparison changes the interpretation: several repeat papers reconstruct earlier work, including a payment repair already developed in a separate follow-up. Reinjection produced a concrete correction and graph extension of a peer-ranking seed, and a conditional-risk counterexample crossing an obedience boundary. Both arms developed related auction information limits. The scientific objective is extension of inquiry: supported ideas developed through generated intermediate results and not observed in the historical or matched repeat record. It is not superiority over repetition, nor a claim that repetition could never discover those ideas. The useful recursive unit is a claim with assumptions and ancestry, not an accepted PDF alone. A focused second-generation follow-up then extends the ranking and risk branches under explicitly matched questions. Its claim audit also identifies an apparent control discovery as a refinement of an original-campaign result.

1 The question: repetition is not recursion

Pathfinder searches for possible connexions between two collections of papers. An abstract-level scan proposes a question for each pair. Selected pairs pass to two researchers, who inspect sources, exchange findings and objections in a ledger, and produce a consolidated note. A verifier may request further work, stop the thread, or recommend a paper. An editor produces a readable account; a recommended paper then passes through author and reviewer stages [1].

A repeat gives a fresh investigation the same original inputs. Reinjection replaces an input by a completed, internally accepted paper from an earlier investigation. Here reinjection occurs at the very end of the pipeline, after paper writing and review, not immediately after editing a research note. It exposes a later team to a compressed argument, including its assumptions, proofs, references and unresolved questions.

Different answers on a second run are not evidence for recursion: fresh research on unchanged inputs also follows different paths. Waiting until repeats appear exhausted would not solve this problem cleanly. There is no known stopping rule that certifies exhaustion, and waiting changes the resources spent before recursion is tried. We therefore used a contemporaneous repeat arm as a control, without claiming saturation [2].

2 The original 10-by-10 space

The saved corpora contain ten Q papers and ten P papers, identified in Appendix B. These are query-returned collections, not pure disciplinary samples: Q includes robotics and astronomical

instrumentation as well as economic mechanisms. The experiment studies this particular retrieved space, not a random sample of science.

The scanner scored feasibility and scientific gain from titles and abstracts, each on a 0–100 scale, and ranked their product. The archive contains 100 scan records, of which 99 were scored successfully. A product threshold of 1500 selected 13 pairs; the human-retained Q1P2, scored 600, made the research shortlist 14. The remaining 86 cells were not investigated in these campaigns. Q1P2 later produced an accepted selective-perception paper. This is evidence of a useful exception to the scan, not an estimate of its false-negative rate.

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
Q1	•	H				•				
Q2										
Q3			•			•			•	•
Q4			•			•			•	•
Q5										
Q6										
Q7	•						•			
Q8								•		
Q9										
Q10										

Table 1: The investigated subset of the original space. Dots mark the 13 threshold selections; H marks the retained human exception. These are selection marks, not successful-result marks.

The original campaign used mixed models and subsequent interventions; its receipts include Claude and GPT stages. A September 23 fresh-context rerun used GPT-6-sol on the same 14 pairs. Their outcomes already showed substantial variation: the original produced five accepted papers and the rerun three. Q4P6 moved from a conditional peer-calibration construction to a PAUSE on strategic shaping; Q8P8 moved from a proposed compliant-grasp study to a conditional wrist-observation obstruction. A changed verdict need not refute an earlier argument: it may reflect a different argument or evidential bar.

3 The repeat–reinjection pilot

Five final papers were frozen as seeds, selected for different argument families rather than solely for recency. Their full text and compiled bibliographies were preserved, with artifact hashes and parentage in the manifest [3].

Seed	Parent	Argument carried forward
S1	O/Q4P6	Shared-comparison calibration and contribution ranking.
S2	O/Q1P2	Room-level calibration transferred to a selective wall gate.
S3	R/Q3P10	Auction surplus with explicit computation costs.
S4	R/Q7P1	True versus expected empirical CVaR, with conditional economic bounds.
S5	R/Q8P8	Limits of wrist-only information for an occluded held object.

Table 2: O denotes the original campaign; R the September 23 rerun. The carbon-mechanism paper was deferred because its source assumptions needed audit. S4 was carried forward for its risk distinction, not as certification of the source mechanism.

The control C repeated the original 14 pairs. The reinjection arm I used 14 distinct Q–seed pairs, excluded each seed’s parent Q, and matched Q frequencies exactly: Q1 appeared three times,

Q3 four, Q4 four, Q7 twice, and Q8 once. Scheduling blocks alternated which arm ran first. No new scan selected treatment pairs after their outcomes were known. In I, local P1–P5 mean S1–S5, not original P1–P5.

Both arms used GPT-6-sol at high reasoning effort, the same frozen engine and prompts, two peer seats, up to four research rounds, and up to three paper rounds. They had separate USD 200 API-equivalent admission caps. Equal caps did not require equal realised expenditure. Each investigation could stop at PAUSE; only DRAFT proceeded to a paper. Only one recursive generation was authorised. Pilot outputs have not been reinjected again.

There were operational interruptions, including authentication and transport failures. Resumption retained stage identities and completed artifacts rather than silently treating retries as new scientific replicates. The final runner and supervisor both report completion; all 28 edits and ten papers report successful builds. All 720 artifacts protected at the last resumption were unchanged, and frozen-input preflight passed. These integrity checks do not establish mathematical validity or complete reference verification.

	O	R	C	I
Investigated pairs	14	14	14	14
Internally accepted papers	5	3	7	3
Other terminal research outcomes	9	11	7	11

Table 3: Descriptive pipeline outcomes. O includes one PAUSE-ON-ITERATE among its nine other outcomes. O and R are historical context, not controlled model comparisons. C and I form the contemporaneous pilot.

Recorded arm costs were USD 52.03 for C and USD 36.52 for I, from 119 and 105 receipts respectively. C has two error receipts. Usage is missing from one of them. I has no error receipts. These are recorded API-equivalent workloads under subscription access, not additional invoices or guaranteed complete metering. They exclude the historical cost of seeds [4]. Supervision is separate: 115 audit logs contain 114 completed-turn usage records, totalling 24,120,478 input tokens, of which 18,601,984 were cached, and 244,038 output tokens. Failed calls and separate probes may lack usage. The pilot took 49.24 hours from staging to completion, including interruption; summed model-call duration is not elapsed wall time.

4 How claims were compared

The comparison unit is a statement with assumptions and an argument, not a paper, title, verdict or topic. We distinguish inherited results, extensions, corrections, new connexions, and proposals still lacking decisive evidence. “Supported” here means that the recorded derivation or finite computation supports the stated conditional claim; it does not mean externally reviewed, experimentally demonstrated, or new in the literature.

The main historical baseline is the union of O and R, including PAUSE ledgers. Close comparisons also use the already-existing Q7 cache-run, September 13 paper follow-ups and September 14 source audit. This is an explicit extension of the initial two-campaign inventory: otherwise the pilot would receive credit for a payment repair already present in the repository. Other model benchmark campaigns are not exhaustively assessed here. Thus novelty means “not found in this specified record”, not “first found anywhere”.

We inspected terminal claims and the relevant proofs, compared ledger entries and earlier supporting notes, and searched 186 historical authored and ledger files for the distinctive constructions. Search matches were checked for substance: a source line numbered 396 is not a 396-board theorem. Absence of a phrase was used to guide reading, not as proof of novelty. The all-board

witness was rerun successfully; the common-flip endpoint calculation was independently checked numerically. This is not a fresh line-by-line proof audit of every manuscript.

The planned label-hidden support assessment was not performed: this reviewer had already seen arm labels. The classifications below are therefore a transparent, unblinded post-hoc assessment. The same distinction between proved conditional results and untested applications is applied to both arms. Appendix A records a disposition for every pilot pair, including PAUSE. Granularity affects counts, so we report result families and their differences rather than manufacture a total “discovery score”.

5 What survives deduplication

5.1 Peer rankings: a real recursive extension, not the first warning about bias

S1’s positive construction uses reports $Y_{er} = X_e Z_{er}$: observers share one realised comparison X_e , with conditionally independent symmetric errors of positive means a_r . Shared-item agreements satisfy $q_{rs} = a_r a_s$. Calibrating observer reliabilities then permits recovery of Bradley–Terry score differences under coverage assumptions.

The historical record already warned that well-fitted comparison scores need not measure true contribution. R/Q3P6, for example, lets a nuisance cue shift the latent score while preserving a Bradley–Terry form. The September 13 S1 exposition also explicitly warns that passing observable checks does not establish conditional independence. Those warnings are inherited.

I/Q1P1 adds a more specific construction. An independent common flip B_e , with constant mean $b > 0$, gives $Y_{er} = X_e B_e Z_{er}$. Agreements still obey $q_{rs} = a_r a_s$, while calibrated edge means become $b \tanh((c_i - c_j)/2)$. A four-vertex path can therefore have the same full report law in two worlds with opposite endpoint rankings. At $b = 1/2$, the true path differences $(4, -3/2, -3/2)$ sum to 1, but their uncorrected transforms sum to approximately -0.26458 .

This supplies a counterexample to the seed’s broad sentence that correlated errors destroy factorisation; it does not refute the seed’s theorem under conditional independence. The new graph results are also substantive: diameter at most two preserves ordinal information within the specified common-flip model; suitable triangles or four-cycles identify its missing scale, while a recorded six-cycle admits ambiguity. These results and a conditional finite-sample interface are not in the inspected historical baseline or C. The general bias warning is old; the explicit same-law construction and graph boundary are the incremental result.

This paper uses S1 deeply. Q1 contributes the distinction between ordinal and numerical predictions and their downstream cost, rather than a new algorithmic performance theorem. Its relevance is principally a correction and extension of the seed, not equal technical contributions from both inputs.

Evidence: O/Q4P6 ledger 28, 32, 37, 41; R/Q3P6 ledger 7, 9; September 13 Q4P6 exposition, limitations; I/Q1P1 ledger 4, 8, 11, 16, 26, 27 and final paper.

5.2 CVaR and obedience: a new conditional connexion

S4 already establishes that the expectation of empirical CVaR can differ from true CVaR and reverse an economically relevant comparison. I/Q4P4 does not discover that bias again. Its increment is to connect the bias to Q4’s state-independent obedience-cycle test, rather than the seed’s resource tax game. The researchers first identify that the paper called Q inside the seed is not the Q assigned to this new pair.

For each finite batch size m , a risky loss is 0 or $2h$ with equal probability. Its expected empirical upper-half CVaR is at most $2h(1 - 2^{-m})$. Put a safe loss $c_m = 2h - h2^{-m}$ between that value and

true CVaR $2h$. In two signals with exchanged action losses, the surrogate supports the prescribed risky actions with zero rewards. True obedience would require both $r(a) - r(b) \geq h2^{-m}$ and its reverse with the same positive margin, which is impossible. The same policy thus crosses an implementability boundary, not merely an objective-value boundary.

This is a supported new connexion relative to the inspected baseline and C, where the risk analysis stays with resource allocation or query cost. It is deliberately narrow: the instance changes with m ; risk values are conditional reduced-form utilities; the impossibility concerns state-independent rewards, not unrestricted state-dependent rewards or CVaR of an entire plan. Cycle-margin and common-reward certificates are useful extensions of standard bounded-error and difference-constraint ideas.

Evidence: R/Q7P1 paper, fixed-batch participation example and value bounds; I/Q4P4 ledger 1, 6, 9–16 and final paper.

5.3 Auction baselines: useful sharpening, substantial inheritance

I/Q1P3 inherits the surplus ceiling 7.5 from S3. Even earlier, O/Q3P3 computed trade-count ceilings, omission/commission losses, and stylised zero-intelligence benchmarks. A claim that reinjection discovered the need for a cheap auction baseline would therefore be wrong.

The new paper supplies a specific conditional no-generated-token fixed-price rule and a price-error counterexample. If its five eligible pairs execute, the rule reaches 7.5. An arbitrarily small upward advised-price error can admit a cost-2 seller who displaces a cost-0.75 seller, losing 1.25 surplus. The latter distinguishes scalar price accuracy from allocation quality. Neither construction establishes behaviour under the actual random scheduler. This is a modest, useful extension of an existing family, not a wholly new auction result or measured benefit of advice.

C/Q3P3's accepted paper also belongs to that family. Its five/six/seven-trade accounting sharpens an older warning that more individually profitable crossings need not increase welfare. I/Q3P2, although PAUSE, adds a rounding-aware lower bound: at least about 44.9% of a reported model's aggregate deficit is attributable to trade count. That numerical interpretation is not found in the historical baseline, but is Q-only: the calibration seed does not supply the bound. Its proposed market calibration lacks run-level data. These should not become three independent discoveries of the same welfare-accounting principle.

Evidence: O/Q3P3 ledger 3–9, 12; R/Q3P10 paper; C/Q3P3 final paper; I/Q1P3 ledger 10, 15, 19, 24; I/Q3P2 ledger 14, 16, 18, 22, 24.

5.4 Auction information limits: a shared family, with a meaningful difference

C/Q3P6 constructs two worlds with the same acting-trader information and the same value multiset, but opposite welfare rankings of crossing and improving a bid. Its final-step continuation is stipulated. Under an equal prior, no local judge can exceed one-half accuracy on the realised better action; the illustrated value of revealing a hidden buyer value is 0.25. This is an explicit auction instantiation beyond O/R's comparator-bias proposals, not evidence about the frequency of such states.

I/Q3P5 imports S5's matched-observation idea into the auction and reaches a related result. Its ledger rejects an initial rent-only example and a symmetry problem, then swaps a value between an active and an already-traded agent and averages over future random invitations. Thus it extends the construction beyond conditioning on a particular next actor. It still needs a stipulated continuation and illustrative prior.

The pair of outputs belongs to one auction-information family, with different conditional examples. The repeat was ACCEPTED and reinjection PAUSE, despite both offering mathematical examples with unresolved empirical relevance. Counting only papers would conceal the shared

finding and this threshold variation. Reinjection adds a refinement here, not exclusive ownership of the information-limit idea.

Evidence: O/R Q3P6 ledgers; S5 final paper; C/Q3P6 ledger 19, 20, 22, 28; I/Q3P5 ledger 4, 15, 18–21.

5.5 Contracts: the clearest repeat-only constructive addition

O/Q4P9 focused on representation and enforcement confounds. R/Q4P9 then proposed path-based contract sufficiency and distinguished feasibility from obedience. C/Q4P9 goes further: its enumeration supplies a reciprocal five-clause tile contract and shared shortest route for every one of the 396 asymmetric boards, giving scores (85, 55) or (75, 65) instead of the outside scores (70, 0) and achieving joint score 140.

We reran the archived enumeration: 314 boards need one Red coverage and 82 need two, reproducing the score counts. This verifies the finite construction under the program’s model, not source-code equivalence with every simulator rule. It is a supported extension beyond the historical proposal and was not found in I. It proves an available mutually improving outcome, not voluntary bargaining or sequential equilibrium. Q supplies the feasibility/obedience distinction; the computation is predominantly about the contract game’s rules.

Evidence: R/Q4P9 ledger 14, 17, 19; C/Q4P9 ledger 20–26, final paper and paper/asymmetric_oracle.py.

5.6 Payment repairs and reward shaping: accepted does not mean new

C/Q7P1’s explicit others-only transfer correction looks new against a short summary of O and R. It is not new against the existing follow-up record. The September 13 paper already has the same correction: its coefficient $\gamma(b_N - 1)/(N - 1)$ equals $\gamma N/(N - 1)^3$, the coefficient in C. The September 14 source audit also supplies direct allocation implementation, participation, non-unique prices and risk-surrogate error control. C’s rediscovery is useful confirmation and exposition, not another independent repair. C/Q7P7 combines that repair with a previously developed carbon charge and forecast-error argument, under restricted optional loads. Its synthesis must not count the same repair again. I/Q7P2’s source contradictions also repeat earlier findings; its calibrated dual-gap proposal remains conditional.

C/Q1P6 is similarly mostly inherited. O/Q1P6 already proves fixed-population telescoping, entry/exit accounting and the loss of ranking information when softmax potentials are pooled. R/Q1P6 supplies critic-offset cancellation and the finite unregularised Bradley–Terry estimate counterexample. Recombining these in an accepted paper is not a new finding that comparisons do not automatically improve learning.

Evidence: September 13 Q7P1 paper, repair equation; September 14 Q7 source audit, static repair; O/Q1P6 ledger 24, 27, 30, 33; R/Q1P6 ledger 4, 9, 18; C/Q7P1, C/Q7P7 and C/Q1P6 final papers.

6 Focused mathematical audit

After the claim comparison, we rederived the principal ranking and risk arguments and checked their numerical witnesses with `notes/check_recursive_claims.py`. This is a separate calculation by the same supervising agent, not blinded external review, literature verification or an empirical test. Archived campaign outputs were left unchanged.

Ranking witnesses and proof boundaries. The path example gives intended endpoint difference 1 and transformed difference -0.264580326901 . For the six-cycle, closure reduces to $S_1 b^4 + S_3 b^2 + S_5 = 0$, where S_k is the degree- k elementary symmetric sum of the six calibrated means. Its

admissible roots are $b = 0.738908509686$ and $b = 0.853764730071$; both close the cycle and reverse the order of vertices 3 and 6, numbering from the first listed edge. These checks support the claimed nonidentification, not just a poor fit. The diameter-two argument follows from the sign identity for an odd, strictly increasing transform; the necessity statement remains restricted to trees. The triangle and four-cycle identities require nondegeneracy. The finite-sample bound follows by propagating log-agreement error, bounding the logit derivative and applying the centred incidence-matrix pseudoinverse. It still assumes a bound on bias relative to the clean target; more labels cannot certify that assumption.

Risk reversal and reward certificates. For loss 0 or 2 with equal probability and upper-half tail risk, the empirical value with k high observations among m is $\min(2, 4k/m)$. Exact binomial summation for $m = 1, \dots, 64$ confirms the paper’s inequalities, including odd batch sizes. The general argument is the all-zero batch event of probability 2^{-m} , so the proof is not limited to those checked sizes. The conditional-versus-ex-ante example also reproduces the reported ex-ante value $3.194 > 3$. The cycle error bound uses two utility errors per signal; the rectangular robust-reward criterion is the usual feasibility condition for difference constraints. No defect was found in these arguments under their stated assumptions. The finite-batch instance changes with m , and neither report-stage incentives nor realised-batch guarantees follow.

Auction arithmetic passes, but the waiting policy needs precision. C/Q3P6’s conditional information value is 0.25. I/Q3P5’s displayed random-invitation regret expression evaluates to 0.112198829651. However, the latter’s supporting construction allows the buyer to cross again if invited while also arguing that selection of the cheap seller within six draws guarantees surplus 2. Early crossing by the buyer can remove that opportunity, so the lower-bound argument needs a more explicit policy than merely avoiding loss-making trades.

A sufficient clarification is to let the informed comparator in state A keep its bid of 2 standing for all six draws, with other agents holding their quotes except for the cheap seller’s profitable crossing. Then the claimed $2[1 - (7/8)^6]$ lower bound holds. The state-B upper bound for a noncrossing first action still follows because a subsequent profitable trade requires the buyer’s reinvitation. Together with the state-A upper bound 0.75 after immediate crossing, these yield the recorded regret bound under this explicit continuation. Thus the information-limit construction is repairable, but its written policy description should be tightened before reuse. We have checked the arithmetic and this logical qualification, not implemented a simulator-equivalent replay or measured the frequency of the constructed history.

7 A focused second generation: extension and rediscovery

On September 28, a prespecified four-thread continuation paired corrected versions of I/Q1P1 and I/Q4P4 with original Q4 and Q1 respectively. Its controls repeated original Q4P6 and Q1P1. Within each block both arms received the same incremental question: target certification under correlated observations, or a costed downstream guarantee from finite-sample risk advice. These are purposive, question-conditioned continuations, not blinded random assignments. The generated inputs were completed papers, with versioned minor corrections; historical sources and outputs were not overwritten.

All four investigations and readable edits finished in 113.50 minutes. Both recursive papers and control Q1P1 were internally accepted; control Q4P6 ended PAUSE. All 34 research calls completed, and 23 Astra audits required no recovery. Recorded workload was 6.90 API-equivalent dollars for controls and 8.70 for recursion, excluding supervision. No arm exhausted its cap. These operational counts do not determine the scientific conclusion.

Ranking: from a common flip to partial identification. The generated ranking seed supplied an identical-report-law reversal under a constant shared flip. The new recursive Q4P1 embeds this in a restricted report-only game: the same proper-score table makes truthful reporting a strict best response in both worlds, although their intended rankings differ. It then permits unknown positive edge-specific flip means. Within this model, an order is forced exactly when the observed edge-sign graph contains a directed path between the relevant vertices. Incomparable vertices admit opposite score orders with the same complete report law. This extends the seed’s constant-rate graph analysis; it does not establish equilibrium selection or downstream performance.

The matched control also constructs truthful-but-inaccurate shared-error examples. Thus the general warning belongs to both branches. The narrower recursive increment is the specified strategic composition and exact edge-specific identification criterion, not the first observation that agreement can mislead.

Risk: from an error box to a priced data certificate. The risk seed supplied deterministic obedience tests for bounded utility errors. Recursive Q1P2 adds simultaneous confidence bounds from realised conditional-loss samples and returns a supporting reward, a certified infeasibility witness, or abstention. Under a strict cycle margin and confidence radius e , its least nonnegative reward cap satisfies

$$R^* \leq R_m^* \leq R^* + 4(n - 1)e,$$

where n is the number of actions and R^* is the true-utility oracle cap. The proof compares longest simple-path potentials after bounded edge inflation. Sampling plus a reward payment can then be priced locally. Concentration and graph feasibility are inherited ingredients; their realised-data use and cap comparison are the incremental construction. Counterfactual sample access, positive margins and a fixed policy remain premises. No system-level competitive ratio follows.

The control’s comparator gap is substantially inherited. Control Q1P1 writes an additive gap between the optimum of a sum of local CVaRs and the optimum aggregate CVaR, gives a tight two-agent example, and adds a conditional holdout/fallback certificate. But O/Q1P1 already derived the objective mismatch and a wrong-optimiser example on September 11, in its composition subsection and Emmy’s F3 note. Its one-sided bound already implies the same comparator accounting after subtracting the system optimum. The new notation and example sharpen an existing result; they are not a new discovery of the mismatch. This revises the preliminary post-run recap.

Audit and evidential limit. We checked the proofs under their stated assumptions and added a deterministic script that enumerates all 543 four-vertex acyclic directed graphs, tests the peer-score identity with exact rational arithmetic, checks 512 utility-box corners against the reward-cap bound, and reproduces both old and new control examples. These checks found no substantive defect, but do not replace external review or empirical validation. The archived reviewers’ minor definition and local-citation corrections are applied only in separate version-2 packages, without a new acceptance claim.

The second generation therefore supplies concrete extensions through the generated intermediates that were not observed in the inspected controls and historical record. It also supplies rediscovery. Neither outcome proves that repetition could never find the recursive claims. The detailed readout records the bounded search scope, sources and remaining application questions.

Evidence: `plans/2026-09-28-extension-readout.md`; `notes/check_extension_claims.py`; `experiments/2026-09-28-recursive-extension/reviewed/`.

8 Deepening an extension through a worked application

After closing the matched comparison, a supervisor-led follow-up instantiated the risk certificate in a synthetic two-signal routing decision. It explicitly prices counterfactual samples, supplies a fallback, caps rewards and charges a bounded surcharge for an incorrect deployment. This permits a conditional expected-cost bound against always using fallback. It is a worked application, not another QP generation or an independently reviewed paper.

For a separated feasible instance, knowing the safe alternative's loss exactly makes a sufficient expected-cost bound smaller than fallback cost. The fixed numerical grid also shows that over-sampling can erase the benefit, while near-boundary instances abstain throughout. Six incorrect deployment certificates occurred in 120000 regime decisions; all were outside their confidence boxes, and their stipulated costs remain in the results. The stronger access regime changes both information and sample cost.

The derivation, declared assumptions and limitations are in the companion note `notes/recursive-risk-deepening.pdf`; code, protocol and hashed results are in `experiments/2026-09-28-risk-deepening/`. This deepens a generated idea by testing its consequences. It does not alter the historical matched comparison or establish real-world usefulness.

9 Conclusions for recursive science

Recursion can change the object of inquiry. The ranking seed becomes an object to challenge, not merely an authority to reuse. The risk seed supplies an ingredient for a different implementability question. These are concrete reasons to continue studying reinjection. They do not show that either result required a generated input or could not have been found by an unobserved repeat.

The objective is extension, not replacement. The repeat arm finds a new finite contract witness and its own information construction. Reinjection contributes distinctive extensions but fewer accepted papers. These counts do not determine whether inquiry was extended. The relevant evidence is a supported new claim, its dependence on a generated intermediate result, and its difference from the historical and control findings. The ranking and risk examples meet that descriptive criterion in the inspected record. A finite repeat sample cannot establish that repetition would never produce them. Shared seeds, selected pairs, heterogeneous research paths and incomplete blinding also rule out a universal causal conclusion.

The historical ledger matters as much as the final paper. PAUSE can contain a sound bound or counterexample whose application is unsupported. Earlier PAUSE work prevents false credit for later acceptance. Separate follow-ups matter too: a paper-only or two-directory inventory would misclassify the payment repair. Ancestry should track claims, assumptions, corrections and unresolved dependencies, not just parent filenames.

Acceptance is a staging decision, not an epistemic guarantee. The seed's overbroad correlation statement survived internal review. The recursive paper usefully challenges it. Conversely, repeating an accepted argument can amplify its authority without adding evidence. All pilot editors also report an arXiv HTTP 406 lookup failure; local generated seeds have missing-public-identifier warnings. Successful builds do not resolve those checks. Reference verification and independent mathematical review remain separate work.

A focused continuation is preferable to waiting for saturation. The focused audit above supports the common-flip and CVaR-cycle results under their restricted models and identifies a

continuation-policy qualification in the recursive auction example. External mathematical review remains open. The September 28 source audit verified the selected seeds’ public references and checked their attribution boundaries, including the source signal-cycle lemma. It is not an exhaustive prior-art search. Versioned seed copies apply the archived reviewers’ minor corrections; the historical outputs are unchanged. A separate auction corrigendum supplies an explicit sufficient waiting policy.

The second-generation protocol selected two generated seeds and two original-input controls, with the same incremental question within each block. It is a purposive second-generation extension test, not a contest in acceptance rates. It does not feed every accepted PDF into an ever-growing pool. Its allocation, admission limits and subsequent run status are recorded separately in `plans/2026-09-28-1322-recursive-extension.md` and `experiments/2026-09-28-recursive-extension/`; no outcome from that follow-up is included in the first-pilot counts above. Its separate results and limitations are assessed in Section 7.

Evidence: `plans/2026-09-28-source-audit.md`; `notes/recursive-auction-corrigendum.tex`.

Daily query intersections, absolute-interest filtering and broader corpus growth are possible later selection strategies, not interventions tested by this pilot. The present result is narrower: a fixed small seed space supports both rediscovery and substantive transformation, and a claim-level memory is needed to tell them apart.

A Complete pilot disposition

The tables give the residual after comparison, not a binary scientific-quality score. A modest extension can be mathematically sound yet insufficient as a paper claim. “New” is relative to the inspected record described in Section 4. The primary evidence for each row is its terminal paper or edited note and ledger; the named O/R antecedents provide the comparison. A means internally accepted paper; P means PAUSE.

Repeat	End	Claim-level disposition
Q4P10	A	Extension: split-prize collision calculation and a bounded priority realisation. The ordered-protocol theorem is established prior work, as the paper acknowledges. O/R pursued paid peer scoring instead. No survival gain was measured.
Q3P9	P	Mostly repeated proposal: immediate settlement is not missing commitment; proposal order and outside options remain confounded. O/R Q3P9 already separate commitment from wording. A role-by-order test is unrun.
Q4P6	P	Conditional extension: bounded payments must distinguish every mixture of profitable deviations, not just each deviation separately. The finite monitor/minimax test and shirking benchmark sharpen O/R’s shaping boundary; no strategic camera or success law was measured.
Q3P10	P	Repeated accounting and untested tariff proposal. The 7.5 ceiling, first-call and scheduler cautions already occur in O Q3P3 and R Q3P10.
Q7P1	A	Rediscovery against the expanded baseline: the same others-only repair and static risk comparison are in the September 13–14 follow-ups. Different presentation is not a new mechanism.
Q4P3	P	Modest extension: two-way separation of peer informativeness and defect truth, plus an F1 filtering inequality. O/R already warn that agreement and copied evidence are insufficient. No review intervention was tested.
Q1P1	P	Extension of R/Q1P1’s rare-tail lower bound from value estimation to an optimisation example with opposite optima. Threshold lifting is known; no new prediction-sensitive query saving is established.

Repeat	End	Claim-level disposition
Q3P3	A	Refinement of old welfare-accounting family, not a new discovery of 7.5 or of inefficient extra crossings. O/Q3P3 already records ceilings and omission/commission. The review intervention remains untested.
Q4P9	A	Supported finite extension: all-board reciprocal contract witness and delivered-coverage identity, beyond R/Q4P9's path-audit proposal. Predominantly P-side, not an incentive theorem.
Q7P7	A	Recombination: earlier payment repair plus O/R's carbon charge, restricted allocation and forecast-error arguments. Conditional synthesis for optional loads; not an additional independent repair.
Q8P8	P	Formal extension: identical Gaussian observations can hide opposite causal directions; a scalar pose-spread statistic can hide free yaw. O/R already question rigidity as a proxy for causal attachment. No hand failure or control improvement was measured.
Q3P6	A	New explicit auction-information example, sharing a family with I/Q3P5. Fixed value multiset, stipulated continuation, no frequency estimate.
Q1P6	A	Largely reconstruction: O/Q1P6 has pooling and entry/exit accounting; R/Q1P6 has critic cancellation and estimator separation.
Q1P2	P	Old first-gate margin plus an elementary unit-audit lower bound $n(1 - 2\delta)$. O/R already isolate missing observations and fallback conditions. The bound is not a physical LiDAR-cost certificate.

Reinject	End	Claim-level disposition
Q4P2	P	Elementary extension of S2: calibrate a maximum over a fixed fitter menu to resist selection. Generic simultaneous coverage and the two-action obedience inequality are not a new measured wall-gate result.
Q3P1	P	Inherited shaping invariance; hypothetical payoff-equivalent feedback. Public-tape ambiguity and a common-anchor example sharpen cautions, but do not establish a feedback effect. Relates to O/R Q3P3 and Q3P6.
Q4P3	P	Modest tariff extension: own-token charges can miss counterparty costs in a fixed-match subgame. S3 already identifies avoided calls. The corrective reward needs metering, funding and enforcement.
Q3P2	P	Supported Q-only numerical extension: the 44.9% deficit bound. Welfare ceilings are inherited. The S2-style calibrated market guarantee has no run-level data and is not a new calibration theorem.
Q7P3	P	Conditional accounting extension: paid learning can exhaust the participation surplus; a geometric sample schedule gives an iteration cap. S3 supplies cost accounting, but Q7 does not posit paid LLM samples.
Q4P4	A	New connexion: finite-batch conditional CVaR reverses an obedience-cycle test. Inherits S4's surrogate bias, adds the specific state-independent implementability boundary and error certificates.
Q1P3	A	Modest extension: completed fixed-price comparator and a price-error/rationing example. S3 and O/Q3P3 already supply the welfare ceiling and cheap-baseline motivation.
Q3P4	P	Conditional instantiation of S4's risk reversal in an auction cross/wait choice. The stipulated continuation and analyst-only fill probability do not explain observed trader behaviour.
Q4P5	P	Extensions under added models: observation-testing bound, measurement-based incentive obstruction, and a coarse-signal oversight counterexample/repair. S5 supplies the information issue. No matched-history handover data or strategic donor is established.

Reinject	End	Claim-level disposition
Q7P2	P	Source corrections are inherited from R/Q7P7 and the September 14 audit. S2-inspired dual-gap calibration is a conditional proposal with no trusted episode labels or nonvacuity demonstration.
Q8P4	P	Elementary S4 risk example and a zero-fall sample calculation requiring 598 independent episodes under its stated target. No closed-loop hardware losses establish a controller result.
Q3P5	P	Supported auction-information construction; shares C/Q3P6’s family but integrates over random invitations. Refinement, not an exclusive reinjection discovery or observed explanation of market losses.
Q1P5	P	Mostly S5’s inherited obstruction, extended with a standard value-of-information identity and marginal set coverage. Practical sensing benefit remains unmeasured.
Q1P1	A	Correction and substantive extension: shared flips can preserve agreement factorisation while reversing rankings; graph structure separates ordinal recovery and scale identification. Seed-driven, with Q1 framing the prediction interface.

B Corpus identity and evidence map

The abbreviated corpus labels below preserve the original ordering. Full titles, abstracts and identifiers are in the root `Q.jsonl` and `P.jsonl`. The query-date prose in the engine note and the earliest saved scan receipts do not agree; corpus files and frozen hashes, rather than that date, identify the experiment here.

ID	Paper (arXiv identifier)
Q1	Learning-Augmented Algorithms (2609.04787)
Q2	Approximately Efficient Multidimensional Bilateral Trade (2609.02872)
Q3	Competitive Market Behavior of LLMs (2609.02580)
Q4	Mechanism Design for Alignment and Control (2609.01595)
Q5	Sensitivity Hot Spot Penalization (2608.30499)
Q6	Facility Location Games Under a Prelocated Facility (2608.30292)
Q7	Mechanism Design with Stochastic Dynamic Stability (2608.29130)
Q8	Aero Hand Open (2608.28578)
Q9	Optomechanical inertial reference for atom interferometry (2608.28418)
Q10	Canadian Galactic Emission Mapper (2608.28159)
P1	Distributed risk-averse optimization via CVaR (2609.04460)
P2	Concept-Guided Exploration (2608.23650)
P3	Adversarial Review (2608.18167)
P4	Uplink-Completion-Triggered Edge-GPU Inference (2608.08330)
P5	Multi-Agent Reinforcement Learning for Base Station Placement (2607.28002)
P6	MARS-RA (2607.27967)
P7	Emission-Forecasting-Based Spatial-Temporal Carbon Response (2607.26560)

ID	Paper (arXiv identifier)
P8	One Hand Watches The Other (2607.22119)
P9	Commitment To Cooperation With Self-Negotiated Contracts (2607.22750)
P10	The Energy Society (2607.14865)

Evidence paths are relative to the repository root. O/pair means `threads/<pair>/`; R/pair means `experiments/2026-09-23-gpt-6-sol-rerun-01/threads/<pair>/`. C and I mean respectively `repeat/threads/<pair>/` and `reinjection/threads/<pair>/` beneath `experiments/2026-09-25-repeat-injection/`. Within each thread, `ledger.jsonl` records numbered entries; `paper/paper.tex` is the final paper when present, and `edited/note.tex` is the readable account. Verdicts do not replace either source of evidence.

The expanded historical baseline additionally includes `experiments/cache-run/threads/Q7P1/`, `experiments/2026-09-13-paper-followups/threads/`, and `experiments/2026-09-14-Q7-source-audit/Q7-repair.tex`. The latter audit and the September 13 Q7P1 paper are essential to the payment-repair deduplication. Their inclusion changes the novelty judgment, not any frozen campaign input or outcome.

The pilot manifest preserves exact seed ancestry, hashes, corpus and prompt hashes, configuration and the 28-entry schedule. Its `outcomes.json`, `arm receipts.jsonl` files and final supervisor audit support the operational totals. These archived records remain unchanged by this review.

References

- [1] Pathfinder. *The Pathfinder engine*. Repository note, `notes/pathfinder-engine.tex`.
- [2] Pathfinder. *Repeat versus reinjection: claim baseline and pilot protocol*. 25 September 2026. `plans/2026-09-25-1623-repeat-versus-reinjection.md`.
- [3] Pathfinder. *Repeat–reinjection pilot archive*. 25–27 September 2026. `experiments/2026-09-25-repeat-injection/`. See Appendix B for the thread-level evidence map.
- [4] Pathfinder. *Repeat versus reinjection: first readout*. 27 September 2026. `plans/2026-09-27-repeat-reinjection-readout.md`. Its initial candidates are superseded by the historical comparison here.